SASKEN

Enabling Adoption of Latest Android Versions Using Project Treble - A Sasken Experience

Whitepaper

Introduction:

Though Google released Oreo more than a year ago, its adoption is still at around 21.5%. Marshmallow and Nougat devices are still the highest while devices with Lollipop are also significantly high.

Users of devices with old versions of Android are not only deprived of new features in the latest versions of Android, those devices are vulnerable to security hacks due to lack of security enhancements done by Google. Moreover, in all likelihood developers have stopped supporting applications on old versions of Android.

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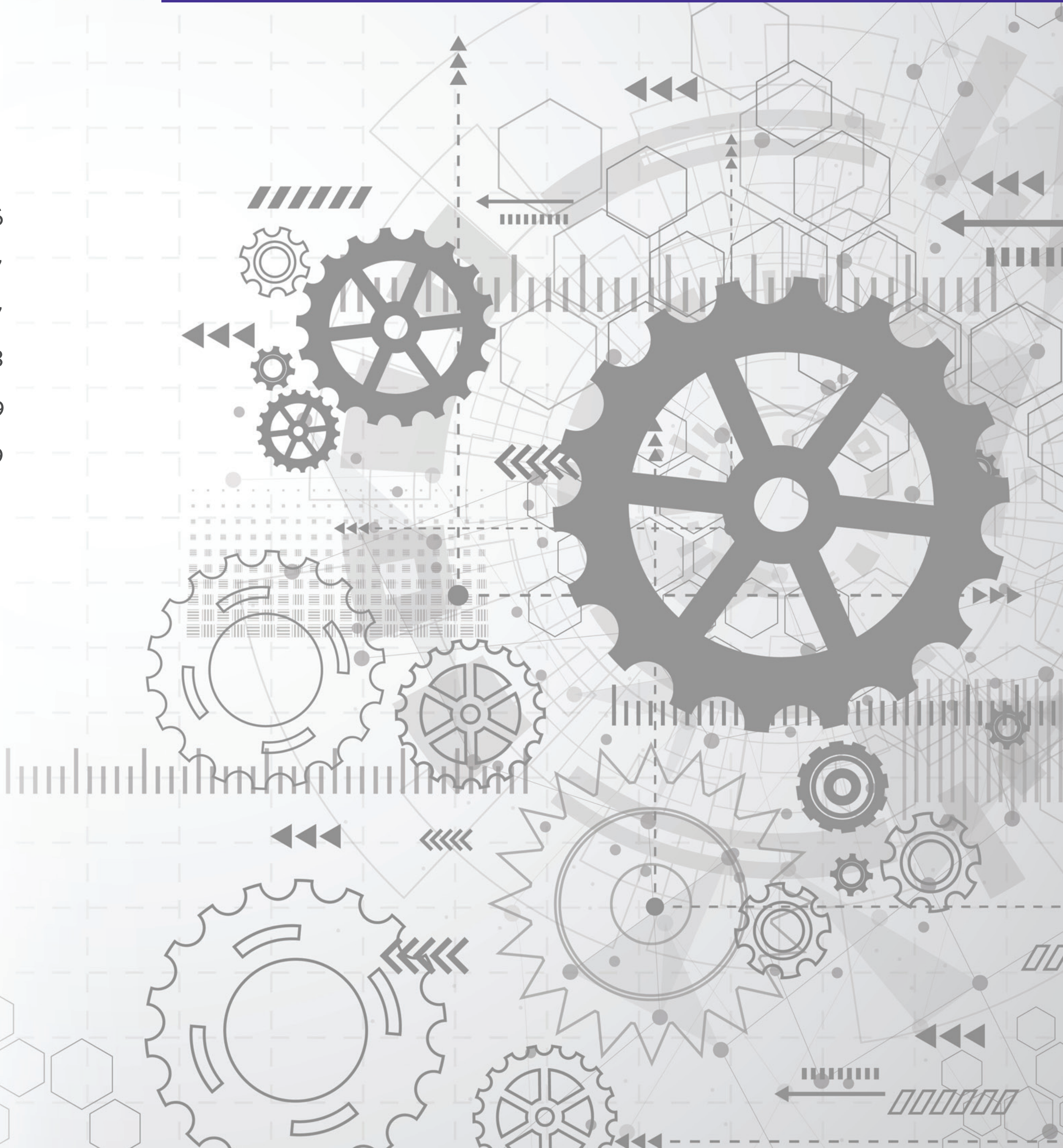
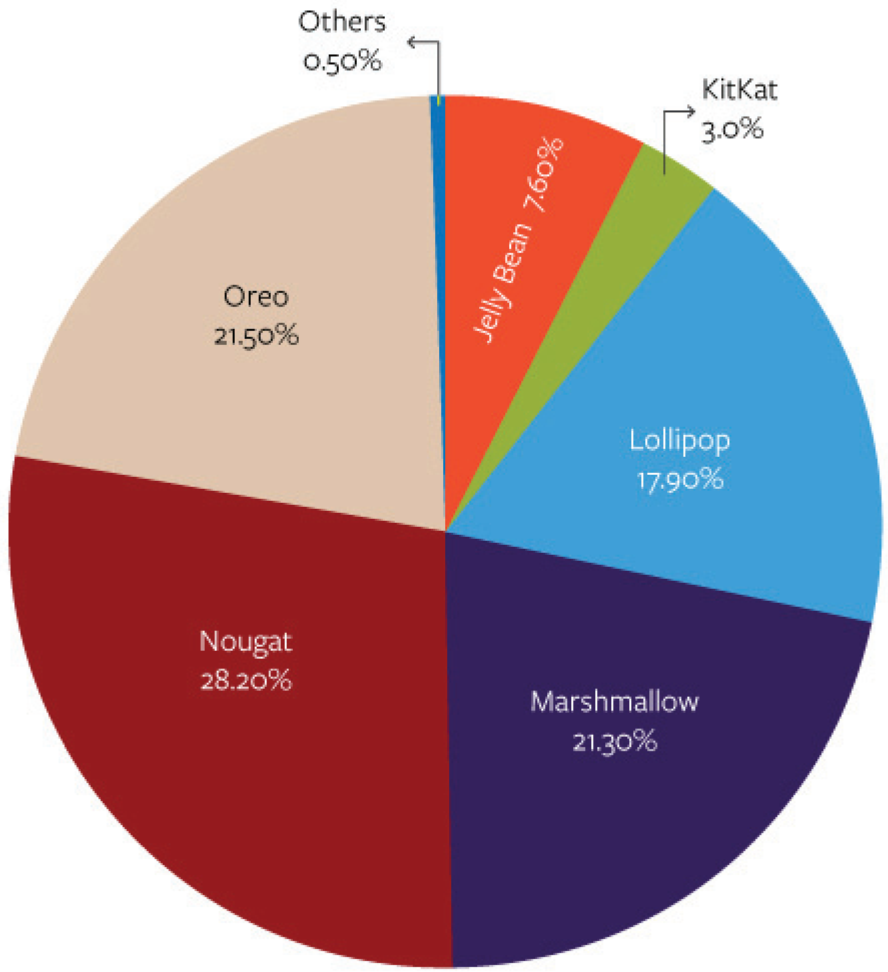
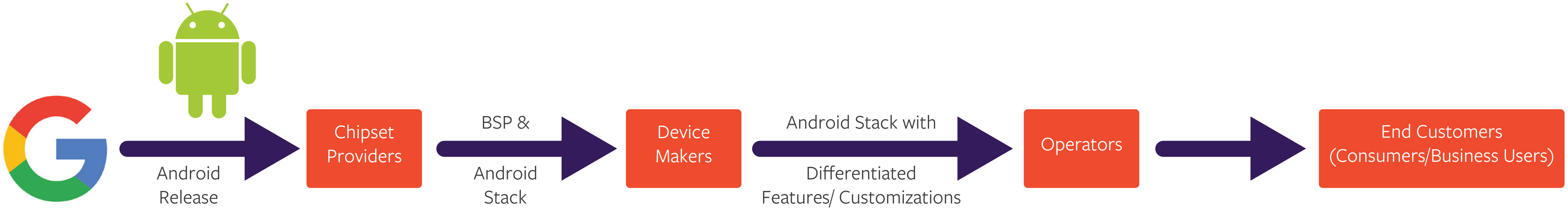


Refer data of Android distribution (as of October 26, 2018) in the following diagram.



Before getting into the reasons for users not getting latest versions of Android, let us look at the life cycle of software for Android device. The following figure shows the software life cycle of Android device.



Chipset vendors take Android stack made available by Google and make it compatible to their chipsets after porting Linux kernel and drivers (called BSP) on the chipset. They then provide this combination to the device makers who make customizations to the Android stack in order to differentiate themselves in market and add features based on operator specifications. Lastly, operators will have to validate the devices to work on their network. This is the process for porting Android on a chipset or to upgrade from one Android version to another.

As long as cost of upgrade is high device makers and chipset vendors will stick to older versions. OEMs will depend on the Android stack tested and provided by chipset vendors with vendor components. Cost can be reduced by shortening the development life cycle. To enable this, Google has introduced Project Treble that ensures that Android framework and middleware are independent of vendor (hardware dependent) layers.

Though users of Android based smartphones always expect the latest version of Android on their phones, chipset vendors, device makers, and operators need to incur the costs for upgrade. Chipset vendors will have to try the new version of Android on their chipsets due to customer demand. But, device makers will be reluctant to incur cost on upgrade because upgrade has to be provided free of cost. Instead, they would prefer to sell a new device with the latest version. Even if they provide an upgrade, quality of upgraded software will be much less than original or previous version(s).

Though Project Treble was introduced in Oreo a year ago, adoption of Oreo version remains at 21.5%. The first time migration to Project Treble is painful since there is architectural change in separation of vendor partition, device maker/carrier customizations, keeping AOSP separate. After the first migration, subsequent upgrades will become very quick.

Advantages of Project Treble

- Device makers can provide upgrades with lesser cost with reduced development life cycle for upgrades by keeping AOSP separate from vendor components

- Application Developers can focus on the latest versions as they need not spend their efforts on supporting all earlier versions as well

- Before Project Treble, any HAL could interact with any driver. With Project Treble, hardware isolation is improved by providing security boundaries between different HAL- Drivers this enhancing security

Sasken's Experience with Project Treble

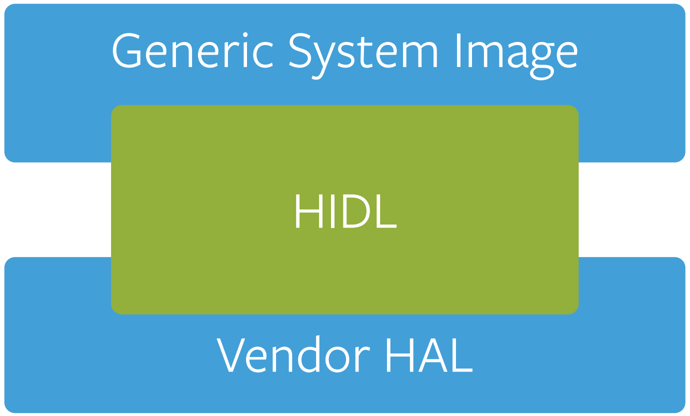
While device makers and chipset vendors can focus on the differentiation that they can bring in, Sasken can support in this first time migration to Project Treble architecture. The company is already supporting device makers and chipsetvendors in migrating to Project Treble in the following areas:

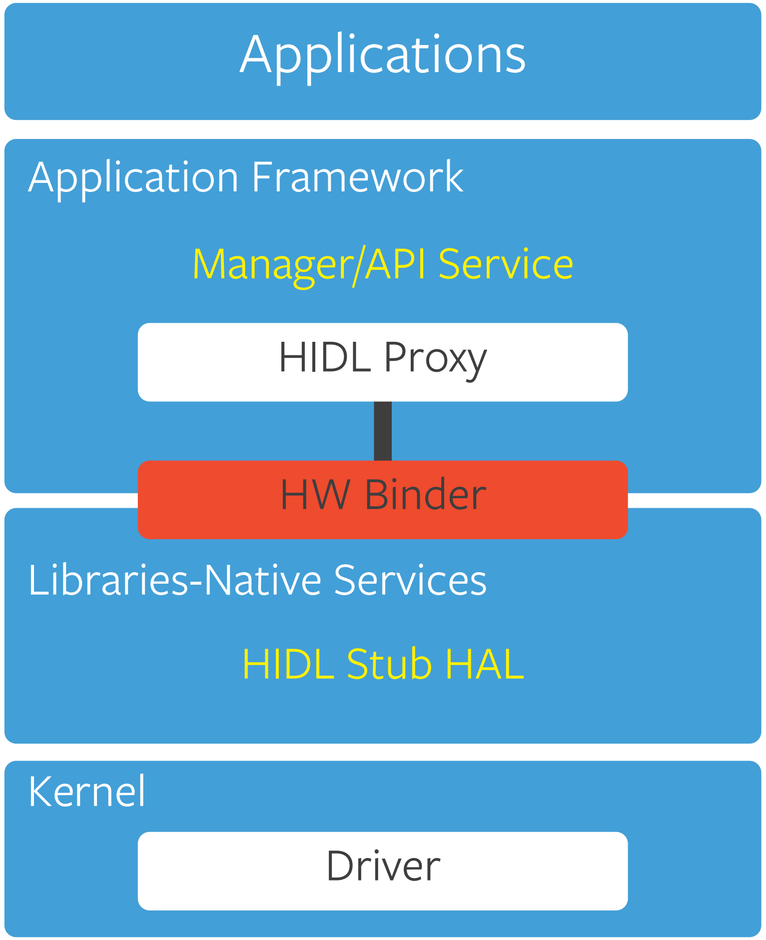
Development of HIDL

Generic System Image is AOSP. Vendor implementation is hardware dependent layer. HIDL ensures that framework can be replaced without need to rebuild HAL. HIDL enables interaction between code that are built independently.

Sasken has implemented HIDL for various peripheral devices with existing HALs and new peripherals that did not have HAL. Architecture for HIDL implementation for a sample peripheral is given on the right.

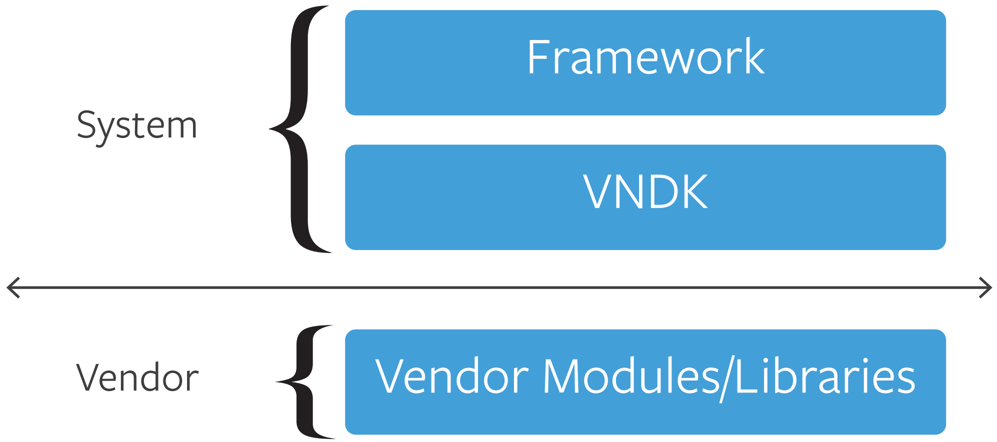
Some of the hardware interfaces for which Sasken had to implement HIDLs were battery, LED, camera.





Adaption to Vendor NDK (VNDK)

VNDK provides a mechanism to implement vendor libraries and HALs and make sure vendor libraries are loaded by vendor processes and not the framework processes. Framework process can access vendor modules only through VNDK.



Sasken has integrated vendor modules/libraries to VNDK.

Though Project Treble has been introduced in Android Oreo version, it is getting officially mandated from Android P onwards. Device makers and chipset vendors will have to put a lot of effort to migrate to the new architecture. While teams of OEMs and chipset vendors can focus on new products, features, Sasken can support in migrating peripheral drivers/HALs to Project Treble architecture along with other phases of development of Android based devices.

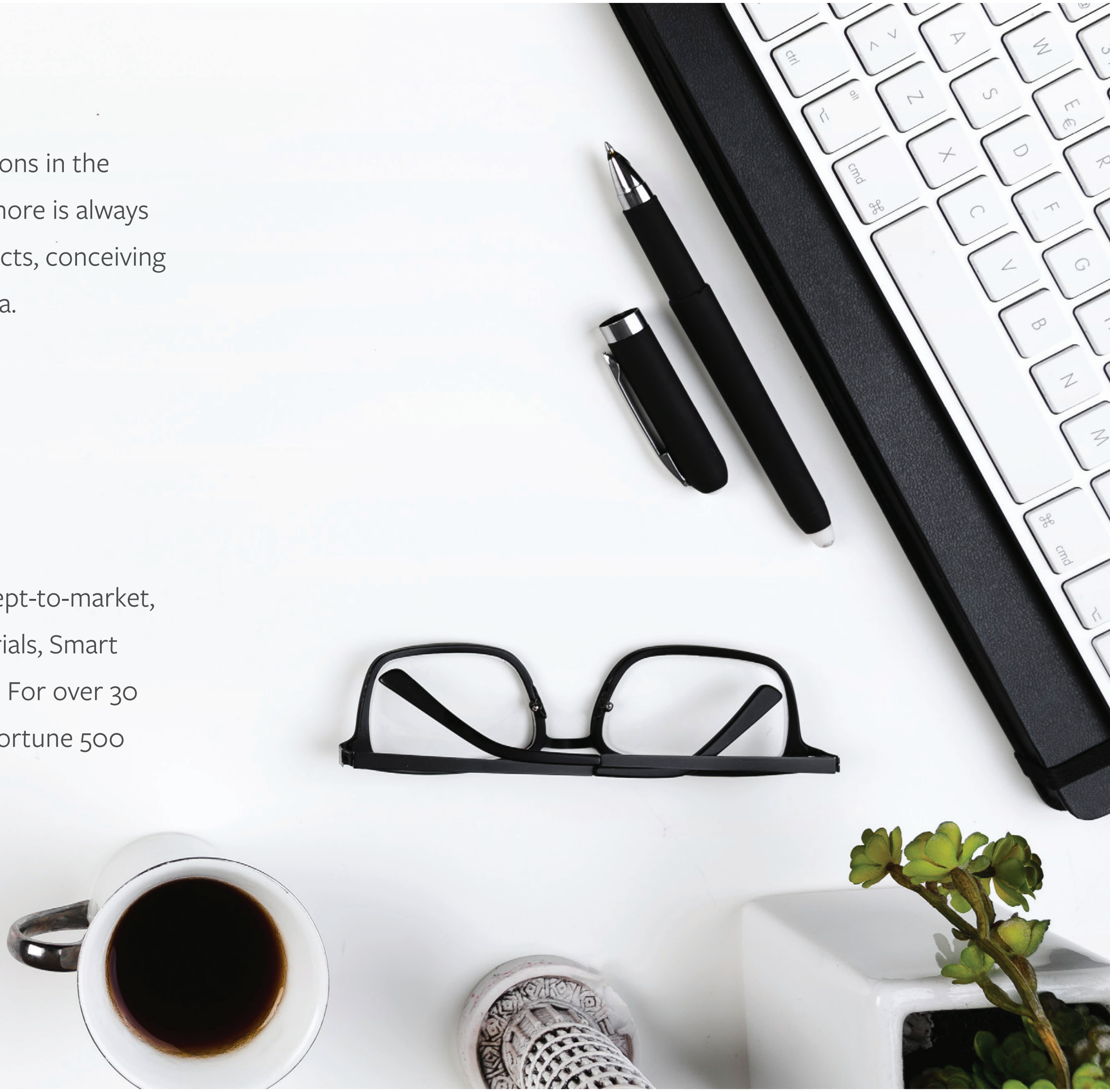
Sasken's expertise in Android spans over a decade and it has helped various OEMs in productizing and launching Android based consumer devices, automotive infotainment systems, and enterprise grade devices (rugged devices). Sasken's involvement is in BSP bringup, multimedia, connectivity, telephony, sensors subsystems, application development, OEM customizations, operator customizations, precertification and certification (Google, Operators). Sasken is involved in customization of Android for In-vehicle-infotainment systems.

About the Author

Kishore has over 20 years of experience in architecting, developing and delivering solutions in the domains of portable devices, consumer electronics, automotive, IoT and wearables. Kishore is always on track with emergence and advancements of technologies, assessing business prospects, conceiving solutions and authoring articles and papers for newspapers, magazines, and social media.

About Sasken

Sasken is a specialist in Product Engineering and Digital Transformation providing concept-to-market, chip-to-cognition R&D services to global leaders in Semiconductor, Automotive, Industrials, Smart Devices & Wearables, Enterprise Grade Devices, SatCom, and Transportation industries. For over 30 years and with multiple patents, Sasken has transformed the businesses of over a 100 Fortune 500 companies, powering over a billion devices through its services and IP.



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Feb 2019